

ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY

LEVEL 5-MODULE 2

UNIT: COMPUTER NETWORK

UNIT CODE: IT/CU/ICTA/CR/03/4/MA

TIME:3 HOURS

PRACTICAL 3

Name**Date.....**

Regno.....sign.....

Instructions

- ✓ *Read all instructions carefully before beginning the practical exercise.*
- ✓ *Ensure that all required equipment and materials are available before starting.*
- ✓ *Observe all laboratory safety rules and regulations throughout the practical session.*
- ✓ *Handle networking devices, cables, and computer equipment with care.*
- ✓ *Use the assigned workstation and equipment only.*
- ✓ *Save all work in the designated locations as instructed.*
 - ✓ *Verify each task before proceeding to the next one.*

Materials provided

- *Terminated Ethernet cables*
- *One switch*
- *One router*
- *Printer and cables*
- *3 desktop computers*

Scenario

You have been employed as a Network Technician in a training institution. The institution requires a small Local Area Network (LAN) to enable users to share files,internet and printers. Your task is to set up the network, configure file sharing, install a printer, and share it across the network.

Task 1: Set Up a Star Topology Network (15 Marks)

Requirements

- 1 Switch
- 3 Computers (PC1, PC2 and PC3)
- UTP Cables(terminated)
- Router

Instructions

- ✓ Connect the three computers to a central switch to form a **Star Topology**
- ✓ **Connect the switch to the router.**
- ✓ **Connect one pc to printer**
- ✓ Power on all devices.
- ✓ Assign the following IP addresses:

Computer	IP Address	Subnet Mask
PC1	192.168.10.1	255.255.255.0
PC2	192.168.10.2	255.255.255.0
PC3	192.168.10.3	255.255.255.0

4. Verify connectivity between all computers using the **ping** command.
5. Demonstrate successful communication among all devices.

Task 2: Create and Share a Folder Containing Files

Instructions

1. On **PC1**, create a folder named **SharedResources**.
2. Create the following files inside the folder:
 - A Microsoft Word document named **StudentList.docx**
 - A Microsoft Excel workbook named **MarksSheet.xlsx**
3. Save both files in the SharedResources folder.
4. Share the folder on the network with appropriate permissions.
5. From PC2 and PC3:
 - Access the shared folder.
 - Open both files successfully.

Task 3: Install a Printer and Printer Drivers

Instructions

1. Connect a printer to PC1.
2. Install the appropriate printer drivers.
3. Confirm that the printer is recognized by the operating system.
4. Print a test page to verify successful installation.

Task 4: Share the Printer Over the Network

Instructions

1. Enable printer sharing on PC1.
2. Assign a printer share name **LabPrinter**.
3. From PC2 and PC3:
 - Search for network printers.
 - Connect to the shared printer.
4. Print a test page from PC2 or PC3 through the shared printer.
5. Verify successful network printing.

